C3: Committed Confidence Calibration

Selection Bias in Confidence-Based Token Commitment in Diffusion Language Models

Mid Submission

Abhiraj Ratna **Ayush SM** **Ojas Kataria** **Parth Sharma**

Abstract

Diffusion language models usually commit several tokens per denoising step, and most decoders choose which by keeping the masked positions with the highest predicted probability. These probabilities are noisy estimates, so the tokens that win could be overconfident(a form of winner's curse). We are measuring this on LLaDA-8B, Dream-7B and a 110M masked diffusion model we trained ourselves, comparing confidence with the actual token(teacher forced). The arguement in our proposal came out to be wrong, that is unequal noise on its own cannot bias the confidence of the committed tokens. So the selection matters only if accuracy at a given confidence depends on context. So we measure the token gap $\Delta(m)$ between confidence and accuracy for a token that beat m candidates, and split it into the model's miscalibration and a field term for context. A 20 document pilot on LLaDA-8B-Base is close to calibrated (ECE 0.006) and the gap does not grow with m: ($\Delta(32) - \Delta(1) = +0.005$, 95% CI $[-0.009, 0.019]$), we should detect effects of about 0.3 points.

1 Introduction

Diffusion language models (dLLMs) generate text by repeatedly denoising a masked sequence. Forward pass predicts a distribution at every masked position, so a step can commit several tokens, and open checkpoints such as LLaDA-8B and Dream-7B have made the question of how many to commit. In most decoders each masked position gets a candidate token and its probability(*confidence*) and the decoder commits the highest values (Chang et al., 2022).

When a decoder keeps the maximum of many estimates, the winner tends to be one whose error was higher than others. Economists and statisticians call the resulting bias the winner's curse (Efron, 2011; Andrews et al., 2024). If it is happening over here, the confidence of a committed token is overstating its accuracy by more when it beats a larger pool, which happens early in generation and for long outputs. Thresholds tuned on that confidence (Wu et al., 2025) would then mean something different in exactly the states where they are applied.

A second, separate account of why parallel decoding loses quality already exists. The model supplies marginals and never the joint probability, so tokens committed together can be make sense individually, but be nonsensical together.

For a committed set S with observed context x_{obs},

$$\mathrm{KL}\Big(p^* \,\Big\|\, \prod_{i\in S} p_{\theta,i}\Big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i\in S} \mathrm{KL}(p_i^* \,\|\, p_{\theta,i})}_{\text{(B) estimation}}. \quad (1)$$

where:

- S is the set of token positions committed (decoded) in parallel in a single step.
- x_{obs} is the observed context: the prompt and all tokens already decoded or unmasked.
- $x_S = (x_i)_{i\in S}$ is the vector of tokens at the committed positions.
- p^* is the true joint distribution over x_S given x_{obs}, i.e. $p^*(x_S \mid x_{\text{obs}})$.
- p_i^* is the true marginal of token i under p^*, i.e. $p^*(x_i \mid x_{\text{obs}})$.
- $p_{\theta,i}$ is the model's predicted marginal for position i, i.e. $p_\theta(x_i \mid x_{\text{obs}})$, with parameters θ. which assumes independence between the tokens.
- $\mathrm{KL}(p \,\|\, q) = \mathbb{E}_p\Big[\log \frac{p}{q}\Big]$ is the Kullback–Leibler divergence.

- $\mathrm{TC}(x_S \mid x_{\mathrm{obs}}) = \mathrm{KL}\big(p^* \,\|\, \prod_{i\in S} p_i^*\big) = \sum_{i\in S} H(p_i^*) - H(p^*)$ is the total correlation: the statistical dependence among the tokens in S given x_{obs}. It is 0 iff the tokens are conditionally independent.

The dependency literature targets (A) and treats (B) as fixed. Selecting on a noisy estimate is one way (B) could be large at exactly the positions a decoder commits.

Contributions. **C1**, a selection-conditional calibration audit on three models: a 110M MDLM we train ourselves, whose training set we control, and LLaDA-8B and Dream-7B. **C2**, separate estimates of the two terms of Eq. 1, exactly in nats on a synthetic JSON probe where p^* is known, and as identified contrasts on real text. **C3**, a recalibration that uses only quantities the decoder already computes and so costs no extra forward passes.

What changed since the proposal. Simulation of proposal's arguement showed it needs correction(§3). The primary endpoints move from the regression coefficient β_z to the committed-token gap and its field term; β_z is kept as a diagnostic, fitted with and without state fixed effects. C3 now runs regardless of C1's outcome. We report conf − agree, so positive means overconfident; the proposal used the opposite sign.

2 Related Work

Masked diffusion language models. Discrete diffusion began with D3PM (Austin et al., 2021a), which introduced the absorbing-state (mask) corruption process as a discrete analogue of DDPM (Ho et al., 2020). MDLM (Sahoo et al., 2024) reduced training to a weighted cross-entropy over masked positions; we follow its recipe for our 110M model. LLaDA and Dream scaled the approach to 7–8B parameters and are the two checkpoints we audit. All of these predict every masked position in one pass, which makes the commit rule the central inference choice.

The commit statistic. Ranking by top-1 probability comes from MaskGIT (Chang et al., 2022) and is the default in LLaDA and Dream. Later work changes the statistic, to margin or entropy, and the cutoff: one global threshold in Fast-dLLM (Wu et al., 2025), and in OSDT a schedule calibrated on a single sequence and reused, justified by confidence trajectories being nearly identical across inputs . These papers report speed and quality, never whether the thresholded values are calibrated. A threshold on a quantity of unknown meaning has no interpretation outside the benchmark it was tuned on.

Stability and calibration of the statistic. KLASS commits tokens whose distribution is stable across steps; SWD proves inter-step instability lower-bounds a token's mutual information with the remaining masked context ;LESS and STDec add persistence and spatial structure. PC-Sampler is the closest correction to ours. It finds an early bias toward trivial tokens and corrects it with a position-aware weight. That bias is positional and content-driven, whereas a selection effect is a property of the ranking operation: it should appear among positions of equal difficulty and scale with field size. We enter PC-Sampler's weight as a competing covariate.

Dependency-aware decoding. Much recent work attributes parallel-decoding error to term (A). DAPD commits independent sets recovered from attention ; DEMASK learns pairwise influence with a total-variation bound ; PUNT selects by conditional-independence tests ; APD uses an autoregressive verifier (Israel et al., 2025); Bansal and Sanghavi (2025) add an auxiliary sampler for approximate joint sampling; and ParallelBench proves subadditivity of conditional total correlation (Kang et al., 2026). All of it takes the marginals as accurate. Tang et al. (2026) observe that current evaluations leave sampler error entangled with model approximation error; C2 is aimed at that entanglement.

Learned policies and theory. Jazbec et al. (2026) learn unmasking policies with reinforcement learning and dParallel distils a model into reaching high certainty faster (Chen et al., 2026b). This is the strongest challenge to our premise, and the answer is that a learned policy can absorb a selection bias without exposing it, so it cannot say whether the bias exists or what it tracks. Cai and Li (2026) bound the iteration complexity of an entropy-sum decoder, taking the predicted distributions as given. Frkovic et al. (2026) find post-hoc remasking on par with confidence unmasking in the short-block greedy setting. MDPO and TRIMS attack the train–inference mismatch from the training side (He et al., 2025; Chen et al., 2026a); our third estimand measures that mismatch. Temper-

Work	Calibration under selection	Separates (A)/(B)
Fast-dLLM; OSDT	no	no
KLASS; SWD; LESS	no	no
PC-Sampler	positional only	no
DAPD; DEMASK; PUNT	no	(A) only
ParallelBench	no	(A) only
Jazbec et al. (2026)	no	no
Tang et al. (2026)	no	identifies
This work	yes	both

Table 1: Position relative to the closest existing work. The last row is what we intend to deliver.

ature in dLLM sampling has itself been studied (Olausson et al., 2026), and a per-step temperature is the obvious competing explanation for any effect we find.

Selection bias and calibration. The winner's curse is formalised through Tweedie's formula by Efron (2011) and for inference on winners by Andrews et al. (2024). Efron's result also yields our Proposition 1: selection does not bias a correct posterior. We use the calibration vocabulary of Guo et al. (2017). Risk-controlled thresholding (Angelopoulos et al., 2024, 2025) would turn a tuned cutoff into a guaranteed error rate, but existing applications cover stopping and acceptance (Schuster et al., 2022; Jazbec et al., 2024) rather than commitment, and it is out of scope.

Viability. Table 1 places this work. No prior study measures calibration conditional on the selection event, and none estimates the two terms of Eq. 1 separately. Every ingredient is available: open checkpoints, an open training recipe, held-out text with ground truth at every position (Penedo et al., 2024), and the statistics of selection. §6 shows the instrument runs on our hardware and that the required sample fits the corpus.

3 Problem Statement

Setup. A *state* s is a document x_0 with a set M of masked positions. One forward pass gives, at each $i \in M$, the top-1 token v_i, its confidence c_i and log-odds $\ell_i = \log \frac{c_i}{1-c_i}$. Under teacher-forced infilling the true token is known, so correctness $a_i = \mathbf{1}\{v_i = x_{0,i}\}$ is observed directly. A pool $P \subseteq M$ of size m is drawn at random, and its positions are ranked $r = 1, \ldots, m$ by ℓ_i. Let $\pi(\ell) = \Pr(a = 1 \mid \ell)$ be the ordinary calibration curve and $\pi(\ell, s) = \Pr(a = 1 \mid \ell, s)$ its context-aware version.

Estimand E1: the committed-token gap. With E_m an average over tokens that win pools of size m,

$$\Delta(m) = E_m[\, c_i - a_i \,], \quad m \in \{1, 2, 4, 8, 16, 32\}. \tag{2}$$

The primary contrast is $\Delta(32) - \Delta(1)$, compared with the standard ECE.

Proposition 1 (Winning adds nothing given confidence and context)**.** *Fix s. Suppose the pairs (ℓ_j, a_j) are independent across $j \in M$ given s, and P is drawn independently of them. Then for the winner i of P,* $\Pr(a_i = 1 \mid \ell_i, s, i = \arg\max_{j \in P} \ell_j) = \pi(\ell_i, s)$.

The result holds however unequal the noise scales are across positions. The proposal's argument, that heteroskedastic noise alone makes field size informative, is therefore wrong. Selection can bias committed confidence only through the second term below.

Proposition 2 (Exact split of the gap)**.** *Under the conditions of Proposition 1,*

$$\Delta(m) = \underbrace{E_m[\sigma(\ell) - \pi(\ell)]}_{\textit{own miscalibration}} + \underbrace{E_m[\pi(\ell) - \pi(\ell, s)]}_{\textit{field term}}. \tag{3}$$

At $m = 1$ winners are random positions, and $\pi(\ell) = E[\pi(\ell, s) \mid \ell]$, so the field term is exactly zero. It is nonzero only when accuracy at fixed confidence depends on context, and winning a large pool shifts which contexts winners come from. The field term is the winner's curse in corrected, testable form.

Estimand E2: the field effect. We estimate the field term as the observed $\Delta(m)$ minus the gap predicted by a calibrator $\hat{\pi}(\ell)$ that sees only the token's own confidence. A context-aware calibrator $\hat{\pi}(\ell, s)$ should then close the remainder.

Estimand E3: trajectory shift. Models are trained on uniformly random masks, but a decoder fills easy positions first. E3 asks whether decoder-visited states are less calibrated than uniform states at equal confidence and matched mask rate.

Diagnostic regressions. The proposal's Eq. 2 and Eq. 3 are kept as diagnostics, fitted on win-

ning tokens:

$$\Pr(a_i = 1) = \sigma\big(\alpha\,[+\alpha_s] + f(\ell_i) + \beta_z z_{r,m}\big), \tag{4}$$

$$\Pr(a_i = 1) = \sigma\Big(\alpha\,[+\alpha_s] + f(\ell_i) + \textstyle\sum_{k\geq 2} \gamma_k \mathbf{1}\{r = k\}\Big),$$

with f a restricted cubic spline, $z_{r,m}$ the expected normal order statistic, and α_s optional state fixed effects. Because the field term lives entirely in differences between states, fixed effects absorb it. So $\beta_z < 0$ without fixed effects and ≈ 0 with them indicates a field effect, while $\beta_z \neq 0$ even with them indicates errors shared within a state, which breaks Proposition 1.

Hypotheses. H1: $\Delta(32) - \Delta(1) > 0$. H2: the field term at $m = 32$ is positive. H3: at equal ℓ and mask rate, decoder states are more overconfident than uniform states. Each is declared only if its 95% interval excludes zero and it exceeds twice the trace-level non-determinism floor. We fix in advance $\delta_{\min} = 0.02$ as the smallest effect worth reporting, used for power.

4 Data

Held-out text. 2,000 documents are reservoir-sampled with a fixed seed from the first 200,000 documents of at least 1,200 characters in the FineWeb-Edu sample-10BT subset (Penedo et al., 2024). A further 500 reserve documents, drawn with a different seed from the same pool, supply the pilot. Documents are split 50/50 by seed: calibrators, splines and C3 are fitted on the first half, and every reported number comes from the second. Regenerating the corpus off-cluster gave a byte-identical manifest.

Contamination. LLaDA and Dream were trained on undisclosed web corpora. We prompt each checkpoint with a document's first 50 tokens, generate 50 greedily, and flag documents whose longest verbatim match with the true continuation reaches 32 tokens. Table 2 shows contamination is negligible. The threshold must be formally pre-registered before flagged documents enter any reported analysis.

Tasks and probe. GSM8K (1,319 test problems; Cobbe et al., 2021) and MBPP (500; Austin et al., 2021b) are formatted as teacher-forced documents with the question as prompt and the reference answer as the masked target; HumanEval (164; Chen et al., 2021) is for end-task evaluation only. Hub revisions and hashes are recorded.

Model (fp16)	Docs	Flagged	Median	Max
LLaDA-8B-Base	1,979	2	1	39
Dream-7B-Base	1,976	1	1	43

Table 2: Contamination screen: longest verbatim run (tokens, of 50) after a 50-token prefix; flagged at ≥ 32.

5 Method and Implementation

Measurement instrument. Given x_0 and a mask, one forward pass records at every masked position v_i, c_i, ℓ_i, a_i, the probability of the true token, entropy, margin, the top-20 ids and probabilities, the change in ℓ_i since the previous step, and the commit rank. *Uniform* masks hide a random fraction t of positions, as in training. *Decoder-induced* masks follow a decoder from a fully masked state, but fill each committed position with the *true* token, so every state a decoder would visit if it were always right is recorded with no knock-on errors. A *matched* uniform arm masks the same number of positions as each decoder state. Traces are stored per (state, position) and pools are drawn at analysis time, so pool sizes and seeds can change without another forward pass.

Estimators. In each state and for each $m \leq |M|$ we draw 20 pools, rank by ℓ with random tie-breaking, and keep the top four ranks. $\Delta(m)$ is averaged within documents first, then across them. $\hat{\pi}(\ell)$ is isotonic; $\hat{\pi}(\ell, s)$ is a logistic model with a spline in ℓ plus the state's median ℓ, mask rate, $\log|M|$ and step; both are cross-fitted by document. We also report the within-state intra-class correlation (ICC) of the residuals $a_i - \hat{\pi}(\ell_i, s)$ as a direct check on shared errors. The regressions use logistic IRLS with fixed effects and a 9-knot spline estimated on the fit half and carried to the report half as a fixed offset. Every interval is a percentile interval from 10,000 document-bootstrap resamples, implemented by reweighting documents and shared across pool sizes, estimands and mask kinds.

The 110M MDLM. A bidirectional transformer with 12 layers, width 768, 12 heads and learned positions, using GPT-2 BPE plus a mask token, trained with the MDLM objective at sequence

length 256 with padding masks and no packing. We used AdamW (learning rate 3×10^{-4}, 2,000 warmup steps, decay to 10%, weight decay 0.02), gradient clipping at 1.0, EMA 0.9999 and fp16 autocast, for 10B tokens (76,293 steps of 131,072 tokens). Evaluation is the NELBO perplexity bound on the 2,000 held-out documents over a fixed grid of 32 diffusion times, with a document-bootstrap interval, beside the exact perplexity of GPT-2 small on the same documents. The same code trains a same-size causal model as the autoregressive reference.

Decoding. One shared loop implements block-wise or whole-sequence decoding with six commit policies: top-k confidence, margin, Fast-dLLM threshold, entropy-bounded, KLASS and PC-Sampler. In teacher-forced mode it reproduces the decoder-induced masks.

6 Progress and Preliminary Results

6.1 The estimator on known truth

Before touching any model we fitted Eq. 4 on synthetic traces with known truth. Under a homoskedastic null it returns $\beta_z = -0.017$. It recovers planted effects of -0.30 and -0.10 as -0.24 and -0.06, so it is conservative. Under a heteroskedastic null, a 5-knot spline leaves curvature that loads onto $z_{r,m}$, giving a spurious $\beta_z = +0.12$ at noise dispersion 1.0; at dispersion 1.5 more knots overshoot to -0.04, which would read as a winner's curse. We fixed 9 knots before any model data, with 5 and 13 as a sensitivity panel, and demoted β_z to a diagnostic.

6.2 T0: four simulated scenarios

Table 4 runs the full analysis on four simulated assumptions. The endpoints separate the worlds. $\Delta(32) - \Delta(1)$ is zero only under calibration, and β_z without fixed effects is strongly negative only when miscalibration depends on context, the true winner's curse. Figure 1 shows why the standard metric is not enough: ECE is 8.3%, but a token that beat eight candidates is overconfident by 12.7%, and a calibrator using only its own confidence explains 8.1%.

Three results depart from the plan. First, the field-term estimator is biased upward: it reads $+0.001$ to $+0.002$ when the truth is zero, and $+0.033$ when the true value is $+0.016$ $[0.004, 0.027]$. Second, fixed effects do not fully clean β_z in the context-dependent scenario

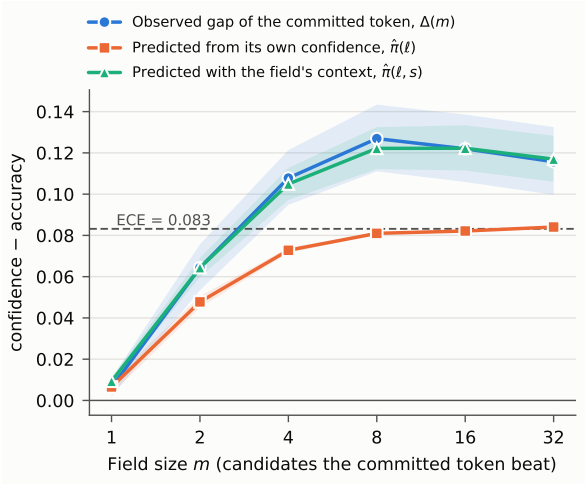


Figure 1: T0, context-dependent scenario. The committed token's gap $\Delta(m)$ grows with field size far past ECE. The own-confidence calibrator $\hat{\pi}(\ell)$ misses the field term; adding context, $\hat{\pi}(\ell, s)$, recovers it.

$(+0.062)$. Third, shared errors produce a *negative* field term with a *positive* β_z, the opposite of the plan's guess. The true field term also does not grow monotonically with m (Figure 4, Appendix A): it peaks at $m = 32$ at low noise but at $m = 4$ for $\sigma \geq 1.5$, and turns negative at $m = 32$.

6.3 Determinism check for fp16

Five repeats of the 20-document pilot were bitwise identical, both in 4-bit on one GPU and in fp16 over two: zero change in ℓ, in agreement and in $\Delta(m)$. The basline therefore is zero.

6.4 T2 pilot on LLaDA-8B-Base

The pilot ran 20 reserve documents in fp16 with uniform masks at five rates and a top-4 teacher-forced trajectory: 1,480 forward passes and 192,088 position rows in 195 seconds on two GPUs. Table 5 and Figure 2 give the results.

Analysis. *The base model is close to calibrated, with no detectable E1 effect.* $\Delta(m)$ stays between 0.5 and 1.1 points at every field size in both arms. The decoder-arm interval on the contrast tops out at $+0.019$, just under $\delta_{\min}$, so a gradient of 2 points or more is already unlikely for this model and corpus. This fits the expectation that base models are near-calibrated on their training distribution, and moves the weight onto the Instruct checkpoints, the task data and E3.

No field effect under decoder masks. The field contrast is -0.0004; the own-confidence calibrator slightly over-predicts the observed gap (0.015

Task	Status	What exists	Remaining
Stage 0 (proposal-era)	done	LLaDA-8B 4-bit, 20 docs, both written gates recorded	superseded by T2
T0 theory, simulations	mostly done	Propositions 1–2; four-scenario table and Fig. 1 from one command	second-person proof check; gate G0
T1 trace pipeline	done	schema v2; uniform, matched and decoder masks; sharded, resumable collection; floor 0	—
T2 pilot, power	mostly done	LLaDA-8B-Base fp16 pilot; gate G1 = full plan	commit split manifest; freeze preregistration
T3 110M model	in progress	v1 trained and evaluated; v2 training on the cluster; AR reference code	v2 evaluation, gate G2, AR run, release
T4 contamination	in progress	screens on both 8B base models	pre-register threshold; time-split set
T5–T7 E1, E2, E3	not started	estimators, calibrators and pilot numbers	full 2,000-document sweep
T8 controls	not started	precision adapters; PC-Sampler weight	precision, position, token-type, temperature runs
T9–T10 C2	not started	fine-tuning hook; $p(\text{true})$ and top-20 in traces	probe generator, exact terms, real-text proxy
T11–T12 C3, decoders	in progress	decode loop, six policies, tasks formatted	recalibration rule; task runs

Table 3: Status by task

Scenario	$\Delta(32)-\Delta(1)$	field term, $m=32$	β_z, no FE	β_z, with FE
Perfectly calibrated in every context	-0.000 $[-0.007, +0.007]$	$+0.001$ $[+0.001, +0.001]$	$+0.010$ $[-0.055, +0.080]$	$+0.005$ $[-0.061, +0.076]$
plan / observed	no / no	$0 / > 0^{\dagger}$	0 / 0	0 / 0
Miscalibrated the same way in every context	$+0.053$ $[+0.044, +0.063]$	$+0.002$ $[+0.002, +0.003]$	-0.006 $[-0.066, +0.056]$	-0.004 $[-0.066, +0.058]$
plan / observed	yes / yes	$0 / > 0^{\dagger}$	0 / 0	0 / 0
Miscalibration depends on context	$+0.109$ $[+0.095, +0.124]$	$+0.033$ $[+0.022, +0.044]$	-0.246 $[-0.305, -0.184]$	$+0.062$ $[+0.001, +0.126]$
plan / observed	yes / yes	$> 0 / > 0$	$< 0 / < 0$	$0 / > 0^{\dagger}$
Errors shared across positions	$+0.042$ $[+0.032, +0.053]$	-0.007 $[-0.009, -0.005]$	$+0.095$ $[+0.034, +0.156]$	$+0.030$ $[-0.033, +0.093]$
plan / observed	yes / yes	$? / < 0$	$< 0 / > 0^{\dagger}$	$\neq 0 / 0^{\dagger}$

Table 4: T0 simulation. Each cell is the estimate with its 95% document-bootstrap interval; the row beneath gives the plan's expected sign and the observed one (†: mismatch).

	Decoder	Uniform
States	1,280	200
ECE	0.006	0.009
$\Delta(1)$	0.006	0.007
$\Delta(32)$	0.011	0.009
$\Delta(32)-\Delta(1)$	$+0.005$	$+0.002$
	$[-0.009, 0.019]$	$[-0.008, 0.012]$
Field(32)−field(1)	-0.0004	$+0.006$
	$[-0.0014, 0.0007]$	$[0.005, 0.007]$
β_z, no FE	$+0.069$	-0.050
	$[0.027, 0.113]$	$[-0.148, 0.050]$
β_z, with FE	$+0.037$	$+0.011$
	$[-0.016, 0.102]$	$[-0.114, 0.140]$
Residual ICC	0.007	0.005
σ_d of contrast	0.032	0.023
Docs for 80% power	20	11

Table 5: T2 pilot, LLaDA-8B-Base fp16, 20 documents. Intervals are 95% document-bootstrap.

against 0.011 at $m = 32$). Under uniform masks the field contrast is $+0.006$ with an interval excluding zero. That is a third of $\delta_{\min}$, but T0 shows the estimator reads high, so we will probe this further to see if the calibrator-refit bootstrap confirms it.

The decoder arm matches T0's shared-error signature. β_z without fixed effects is positive with an interval excluding zero, the field term is slightly negative, and the residual ICC is small but nonzero ($[0.004, 0.010]$). This is row four of Table 4. The rank profile agrees: at equal confidence, rank-4 positions are less accurate than the winner ($\gamma_4 = -0.079$, $[-0.149, -0.002]$). This is exploratory: one of three γ_k, uncorrected, on 20 documents.

Committed tokens. Tokens the top-4 decoder actually committed are overconfident by 1.0 to 2.0 points when 32 to 128 positions remained masked, with intervals that include zero.

Precision may matter. The proposal-era pilot used 4-bit weights and different documents. It found committed tokens overconfident by 3 to 9 points and a field contrast of $+0.055$, about ten times the fp16 values.

Power. With $\sigma_d = 0.032$, a 1,000-document report half gives a standard error of about 0.001 on the contrast, so effects near 0.3 points are detectable. Gate G1 resolves to running the full plan ($N = 20 \leq 1{,}000$).

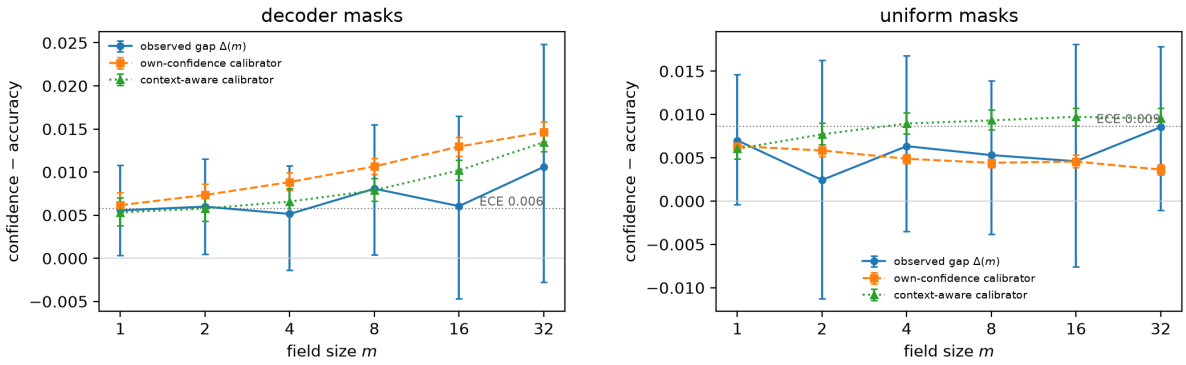


Figure 2: T2 pilot: observed $\Delta(m)$ with the gaps predicted by the own-confidence and context-aware calibrators, under decoder-induced (left) and uniform (right) masks. The dotted line is ECE.

6.5 The 110M model

The v1 model's perplexity bound on the 2,000 held-out documents is 38.4 $[37.7, 39.2]$, against an exact 28.5 $[28.0, 29.0]$ for GPT-2 small (124M). A gap is expected: MDLM gives only an upper bound, and GPT-2 was trained on different data with far more compute. The fair comparison is a same-size autoregressive model on our data. Samples are locally fluent but globally incoherent, as expected at this scale.

7 Challenges and Deviations

Identification. The largest change is conceptual. Proposition 1 shows the proposal's heteroskedasticity argument was wrong, and T0 showed that β_z with fixed effects cannot see the field term. We found this in simulation before spending GPU time and reframed the endpoints (§3).

Estimator weaknesses. The spline control can create spurious β_z up to $+0.12$; the field-term estimator reads about twice the true value; and E2 intervals currently treat the fitted calibrators as fixed. Knots are now fixed in advance.

8 Timeline for the Rest of the Project

The C1 sweep is the critical path; the probe is built in parallel so that C2 does not wait for it.

1. **By 4 Oct:** the v2 model evaluated and gate G2 recorded; split manifest, contamination threshold and analysis configuration frozen as the pre-registration; gate G0 sign-off on the reframing.
2. **By 11 Oct:** all 2,000 documents collected for the 110M model (fp32) and both 8B base models (fp16), with uniform, matched and decoder masks; Instruct checkpoints on GSM8K and MBPP.
3. **By 14 Oct:** primary analysis on the report half with the knot sensitivity panel, with and without flagged documents, and with calibrator-refit intervals for E2. The team decides here how far the field effect should shape C3.
4. **By 18 Oct:** controls as rows of one ablation table: 4-bit against fp16 on 200 documents, the PC-Sampler weight and position covariates, token type, and a per-step temperature.
5. **By 20 Oct:** C2 on the JSON probe, with a dependency knob $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ and $k \in \{1, 2, 4, 8\}$ tokens per step on the fine-tuned 110M model, requiring (A)+(B) to match the directly computed KL to 10^{-6} nats; then the parallel-versus-sequential disagreement proxy on 300 real documents.
6. **By 25 Oct:** C3, a context-aware threshold, against the six decoders on GSM8K and

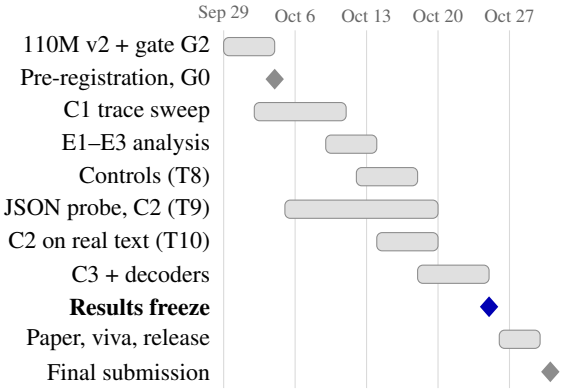


Figure 3: Remaining work, 29 September to 31 October 2026.

MBPP at matched tokens per step. A null is reported as a null.

7. **26–30 Oct:** paper, viva slides, release of traces and checkpoint.

More. A third model, longer outputs, block-wise decoding and calibrator data-efficiency ablations.

References

Isaiah Andrews, Toru Kitagawa, and Adam McCloskey. 2024. Inference on winners. *Quarterly Journal of Economics*, 139(1):305–358.

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. 2025. Learn then test: Calibrating predictive algorithms to achieve risk control. *Annals of Applied Statistics*, 19(2):1641–1662.

Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. 2024. Conformal risk control. In *International Conference on Learning Representations*.

Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021a. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*.

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, et al. 2021b. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.

Parikshit Bansal and Sujay Sanghavi. 2025. Enabling approximate joint sampling in diffusion LMs. *arXiv preprint arXiv:2509.22738*.

Changxiao Cai and Gen Li. 2026. Confidence-based decoding is provably efficient for diffusion language models. *arXiv preprint arXiv:2603.22248*.

Huiwen Chang, Han Zhang, Lu Jiang, Ce Liu, and William T. Freeman. 2022. MaskGIT: Masked generative image transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11315–11325.

Lingjie Chen, Ruizhong Qiu, Yuyu Fan, Yanjun Zhao, and Hanghang Tong. 2026a. TRIMS: Trajectory-ranked instruction masked supervision for diffusion language models. *arXiv preprint arXiv:2604.00666*.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, et al. 2021. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*.

Zigeng Chen, Gongfan Fang, Xinyin Ma, Ruonan Yu, and Xinchao Wang. 2026b. dParallel: Learnable parallel decoding for dLLMs. In *International Conference on Learning Representations*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.

Bradley Efron. 2011. Tweedie's formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614.

Stipe Frkovic, Metod Jazbec, Dan Zhang, Christian A. Naesseth, Ilija Bogunovic, and Eric Nalisnick. 2026. Re-evaluating confidence remasking in masked diffusion language models. *arXiv preprint arXiv:2606.12232*.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*.

Haoyu He, Katrin Renz, Yong Cao, and Andreas Geiger. 2025. MDPO: Overcoming the training-inference divide of masked diffusion language models. *arXiv preprint arXiv:2508.13148*.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*.

Daniel Israel, Guy Van den Broeck, and Aditya Grover. 2025. Accelerating diffusion LLMs via adaptive parallel decoding. In *Advances in Neural Information Processing Systems*.

Metod Jazbec, Theo X. Olausson, Louis Béthune, Pierre Ablin, Michael Kirchhof, João Monteiro, Victor Turrisi, Jason Ramapuram, and Marco Cuturi. 2026. Learning unmasking policies for diffusion language models. In *Proceedings of the 43rd International Conference on Machine Learning*.

Metod Jazbec, Alexander Timans, Tin Hadži Veljković, Kaspar Sakmann, Dan Zhang, Christian A. Naesseth, and Eric Nalisnick. 2024. Fast yet safe: Early-exiting with risk control. In *Advances in Neural Information Processing Systems*.

Wonjun Kang, Kevin Galim, Seunghyuk Oh, Minjae Lee, Yuchen Zeng, Shuibai Zhang, Coleman Hooper, Yuezhou Hu, Hyung Il Koo, Nam Ik Cho, and Kangwook Lee. 2026. ParallelBench: Understanding the trade-offs of parallel decoding in diffusion LLMs. In *International Conference on Learning Representations*.

Theo X. Olausson, Metod Jazbec, Xi Wang, Armando Solar-Lezama, Christian A. Naesseth, Stephan Mandt, and Eric Nalisnick. 2026. A tale of two temperatures: Simple, efficient, and diverse sampling from diffusion language models. *arXiv preprint arXiv:2604.09921*.

Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. 2024. The FineWeb datasets: Decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*.

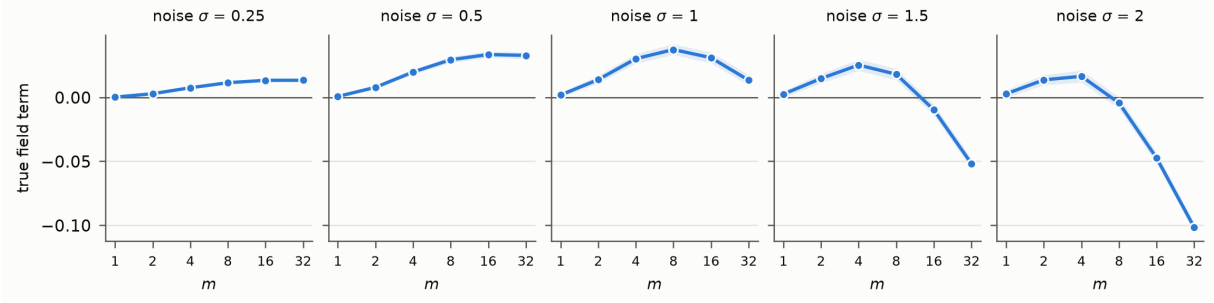
Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T. Chiu, Alexander Rush, and Volodymyr Kuleshov. 2024. Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems*.

Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and Donald Metzler. 2022. Confident adaptive language modeling. In *Advances in Neural Information Processing Systems*.

Luhan Tang, Longxuan Yu, Shaorong Zhang, and Greg Ver Steeg. 2026. Is your diffusion sampler actually correct? A sampler-centric evaluation of discrete diffusion language models. In *Proceedings of the 43rd International Conference on Machine Learning*.

Chengyue Wu, Hao Zhang, Shuchen Xue, Zhijian Liu, Shizhe Diao, Ligeng Zhu, Ping Luo, Song Han, and Enze Xie. 2025. Fast-dLLM: Training-free acceleration of diffusion LLM by enabling KV cache and parallel decoding. *arXiv preprint arXiv:2505.22618*.

A Additional Figures



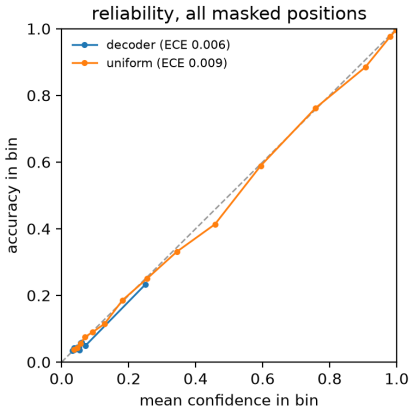


Figure 4: Top: true field term against field size at five noise levels (T0). It grows with m at low noise but peaks at $m = 4$ to 8 and turns negative at high noise. Bottom: T2 pilot reliability over all masked positions. Decoder-state positions sit at low confidence because most decoder states are heavily masked.

B Proofs

Setting. Let s be a state with masked positions M, $\ell_{-i} := (\ell_j)_{j \in M \setminus \{i\}}$, and $\pi(\ell_i, s) := \Pr(a_i{=}1 \mid \ell_i, s)$ (it may also depend on i; we suppress the index). A pool $P \subseteq M$ is drawn independently of $(\ell_j, a_j)_{j \in M}$ given s, ties are broken by independent randomness U, and W_i is the event that $i \in P$ wins the pool, $i = \arg\max_{j \in P} \ell_j$.

Proposition 1. *Suppose that, given s, each pair (ℓ_i, a_i) is independent of ℓ_{-i} (e.g. the pairs are independent across positions). Then, almost surely on W_i,*

$$\Pr(a_i{=}1 \mid \ell_i, s, W_i) = \Pr(a_i{=}1 \mid \ell_i, s) = \pi(\ell_i, s).$$

Proof. Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$. Given s, $(\ell_i, a_i) \perp \ell_{-i}$ and $(P, U) \perp (\ell_j, a_j)_{j \in M}$, so $(\ell_i, a_i) \perp Y \mid s$, and by weak union $a_i \perp Y \mid Z$, i.e. $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$ a.s. Since W_i is determined by (Y, Z), for every bounded measurable h,

$$\begin{aligned} \mathbb{E}\big[a_i\, h(Z)\, \mathbf{1}_{W_i}\big] &= \mathbb{E}\big[\mathbb{E}[a_i \mid Y, Z]\, h(Z)\, \mathbf{1}_{W_i}\big] \\ &= \mathbb{E}\big[\pi(\ell_i, s)\, h(Z)\, \mathbf{1}_{W_i}\big]. \end{aligned} \tag{5}$$

Dividing by $\Pr(W_i) > 0$ shows that $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s. □

Proposition 2. *Let $\mathbb{E}_m$ average over winners of size-m pools (states may be weighted by any function of s), under the hypothesis of Proposition 1. With (ℓ, a) the winner's pair, $\Delta(m) := \mathbb{E}_m[\sigma(\ell) - a]$ and $\pi(\ell) := \Pr_1(a{=}1 \mid \ell)$ the ordinary calibration curve (no selection, $m = 1$),*

$$\Delta(m) = \underbrace{\mathbb{E}_m\big[\sigma(\ell) - \pi(\ell)\big]}_{\text{ordinary miscalibration}} + \underbrace{\mathbb{E}_m\big[\pi(\ell) - \pi(\ell, s)\big]}_{\text{field term}},$$

and the field term is exactly zero at $m = 1$.

Proof. Exactly one W_i holds. Applying (5) with $h \equiv 1$, summing over i and averaging over s,

$$\begin{aligned} \mathbb{E}_m[a] &= \textstyle\sum_i \mathbb{E}_m\big[a_i \mathbf{1}_{W_i}\big] \\ &= \textstyle\sum_i \mathbb{E}_m\big[\pi(\ell_i, s) \mathbf{1}_{W_i}\big] = \mathbb{E}_m[\pi(\ell, s)], \end{aligned}$$

so $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$; add and subtract $\pi(\ell)$. At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ by the display, so the field term is 0. □